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A System for Automatic Normalization of *Surzhyk* and Dialectal Forms of the Ukrainian Language

«Lexicon»: a system that translates four non-standard varieties of Ukrainian into the literary norm.

Denys Koval
Student, group ШІ-42

Oleksii Syvokon
Supervisor

Computer Science · 122
Dept. of Artificial Intelligence Systems



— PROBLEM

Four non-standard varieties of Ukrainian



Surzhyk

Ukrainian-Russian mixed speech,
common across the country



Hutsul

Archaic Carpathian dialect with Polish
and Romanian influences



Boiko

Mountain dialect with Polish and Old
Slavic influences



Transcarpathian

Shaped by Hungarian, Slovak and
Romanian

Open data is scarce: one small parallel corpus exists for Hutsul, none for Boiko or Transcarpathian. LanguageTool and similar tools treat these varieties as spelling errors; Vuyko Mistral covers Hutsul only.

Goal: translate all four into literary Ukrainian automatically, and check whether small fine-tuned models can compete with commercial LLMs here.

Goal and scope of the work

Goal

Develop a system that automatically normalizes surzhyk and three regional dialects (Hutsul, Boiko, Transcarpathian) to the literary standard of modern Ukrainian, using neural models and low-resource data techniques.

Object

The process of automatic normalization of non-standard forms of modern Ukrainian - surzhyk and regional dialects.

Subject

Methods of building parallel datasets for low-resource language varieties, and methods of training translation models and LLMs for normalization.

Scientific novelty

First single system covering surzhyk + three dialects; first public synthetic parallel datasets for Boiko and Transcarpathian; comparative evaluation of a translation model vs. an LLM on multi-dialect normalization.

Practical value

The «Lexicon» web application: normalizes four varieties of Ukrainian to the literary standard and supports voice input.

— RELATED WORK

Existing tools

Capability	Vuyko Mistral	Spivavtor / UA-GEC	LanguageTool	Commercial LLMs	«Lexicon» (ours)
Surzhyk normalization	—	± as errors	± as errors	± weak zero-shot	✓
Regional dialects	± Hutsul only	—	—	± weak zero-shot	✓ three dialects
Open parallel dataset	✓ Hutsul	± GEC edits	—	—	✓ 125,615 pairs
Single model, many varieties	—	—	—	± generalist	✓
Voice input	—	—	—	± poor on dialects	✓ ASR module

Closest related work: Vuyko Mistral (2025), LLM adaptation for Hutsul translation; Spivavtor (2024), instruction-tuned Ukrainian text editing.

— APPROACH

Approach

GIVEN

9,853 manual Hutsul pairs ·
normative sentences from
Spivavtor · scanned dialect &
surzhyk dictionaries (PDF)



FIND

A single system that normalizes
text from any of the four varieties
to the literary standard



METHODS

Synthetic data expansion (rules +
LLM) · fine-tuning translation
models · prompt engineering



RESULT

«Lexicon» web app: pick a variety,
type or speak, get normalized
literary Ukrainian

*We generate pairs in the direction **norm** → **non-standard**: the literary side comes from a verified corpus, so it stays correct.*

Corpus synthesis: two pipelines

Rule-based - surzhyk **20,911 pairs**

- Phase 1: phrase-level substitution via regular expressions
- Phase 2: word-level substitution with **pymorphy2** lemmatization, preserving case, gender and number agreement
- Deterministic and dictionary-driven; needs no manual annotation

LLM-generated - dialects **108,812 pairs**

- TF-IDF cosine similarity selects the 10 nearest dictionary entries per sentence
- **Mistral-Small-24B** (4-bit, local inference), batches of 5 sentences, JSON output
- Quality control: double parsing, filtering of incomplete outputs, deduplication

Train/validation split 90/10. The **test set is fully manual**: 200 pairs, 50 per variety, with no overlap with training data.

— DATA

Corpus: *125,615* training pairs

Variety	Manual	LLM-generated	Rule-based	Train	Validation	Test (manual)
Hutsul	9,853	19,841	-	26,725	2,969	50
Boiko	-	29,754	-	26,778	2,976	50
Transcarpathian	-	29,605	-	26,644	2,961	50
Surzhyk	-	29,612	20,911	45,471	5,052	50
Total	9,853	108,812	20,911	125,615	13,756	200

Only Hutsul comes with manual training data.

— MODELS

Two models

umt5-base encoder-decoder · $\approx 300\text{M}$ params

- **Full fine-tuning**, one model for all four varieties
- Variety prefix in the input: *"Translate from Hutsul: ..."*
- 40 epochs · lr 5×10^{-5} · label smoothing 0.1 · early stopping (patience 5)
- Decoding: beam search

MamayLM decoder-only · Gemma-3-4B · $\approx 4\text{B}$ params

- **QLoRA**: base weights frozen in 4-bit, only low-rank adapters trained (r=16, $\alpha=32$: $\approx 13\text{M}$ weights, 300× fewer)
- Variety marker in the system role of the chat template
- 3 epochs · lr 2×10^{-4} · Unsloth acceleration
- Decoding: sampling, $\tau=0.1$, top-k 25

We also tried mBART, ByT5 and Lapa LLM; umt5-base and MamayLM scored best in the initial Hutsul-only baseline runs.

Training umt5-base



314k steps over the multidialect corpus; training loss settles near 1.9, validation BLEU near 71.7.

We kept umt5-base as the main model: it is 13× smaller than MamayLM, scores higher on three of the four varieties, and responds faster. The web app runs on it.

— RESULTS

chrF++ on the manual test set

Model	Surzhyk	Hutsul	Boiko	Transcarpathian
umt5-base (ours)	93.58	86.90	54.78	27.45
MamayLM (ours)	91.22	74.02	66.82	20.93
GPT-4o (zero-shot)	88.97	64.89	-	65.39
Gemini 2.0 Flash (zero-shot)	86.67	65.34	74.57	68.16
Mistral Large (zero-shot)	85.79	63.18	70.91	63.65
MamayLM before fine-tuning	65.58	16.73	58.73	27.90

— RESULTS

Hypotheses and outcomes

H1 Fine-tuned model reaches chrF++ ≥ 85 on surzhyk

CONFIRMED

Both of our models passed the threshold: umt5-base - **93.58**, MamayLM - **91.22**.

H2 Fine-tuned models vs commercial LLMs

CONFIRMED*

+6.91 chrF++ on surzhyk and +21.56 on Hutsul over the strongest zero-shot baseline. Fine-tuning lifted MamayLM itself by +25.64 (surzhyk) and +57.29 (Hutsul).

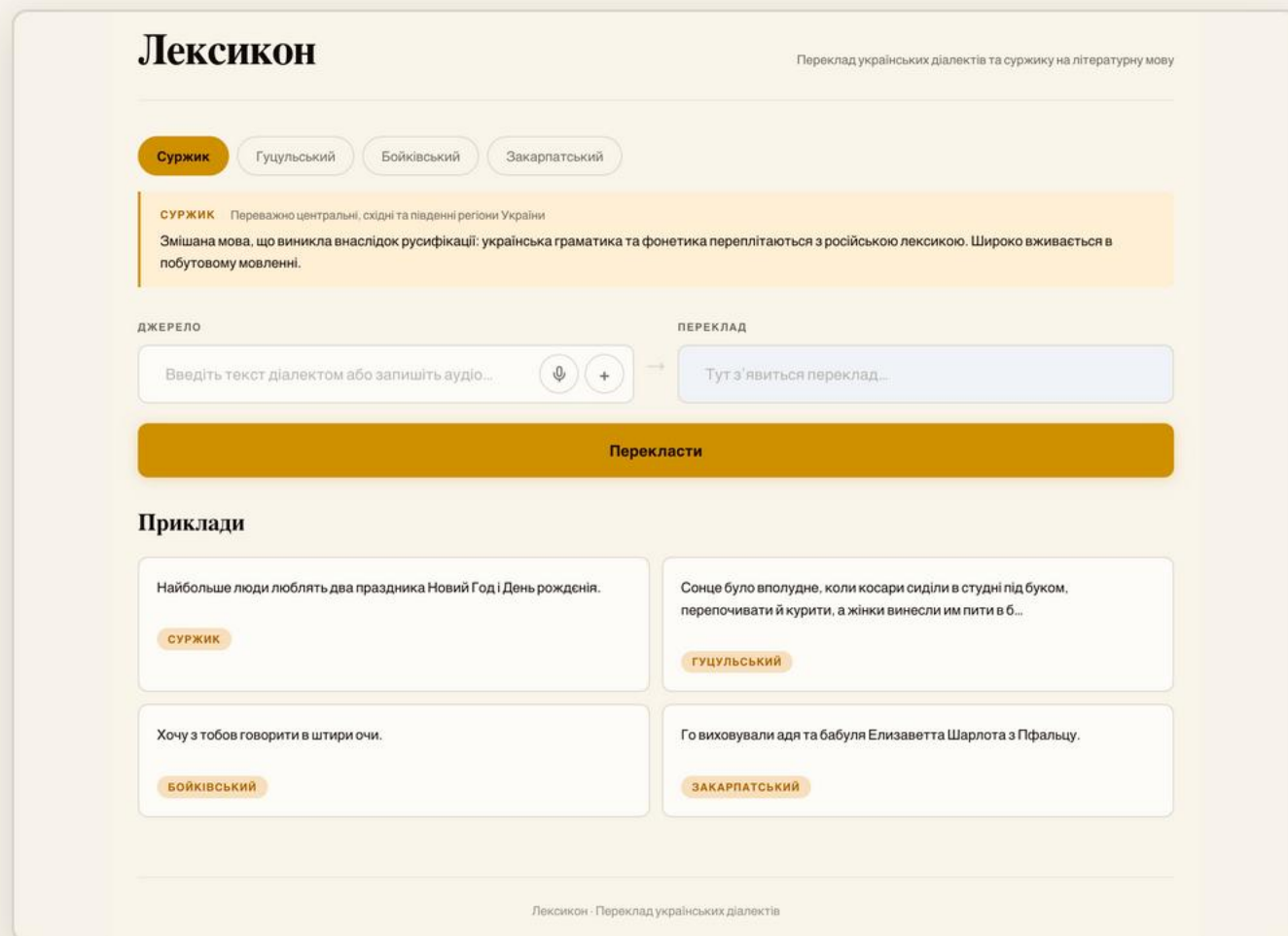
*Holds for varieties with quality training data.

H3 Average chrF++ ≥ 75 across all four varieties

NOT CONFIRMED

Average: 68.69. Boiko (66.82) and Transcarpathian (27.45) train on synthetic data only; the commercial models stay ahead there.

The «Lexicon» web app



Stack

NiceGUI + FastAPI; the interface is ≈ 250 lines of Python.

Normalizer

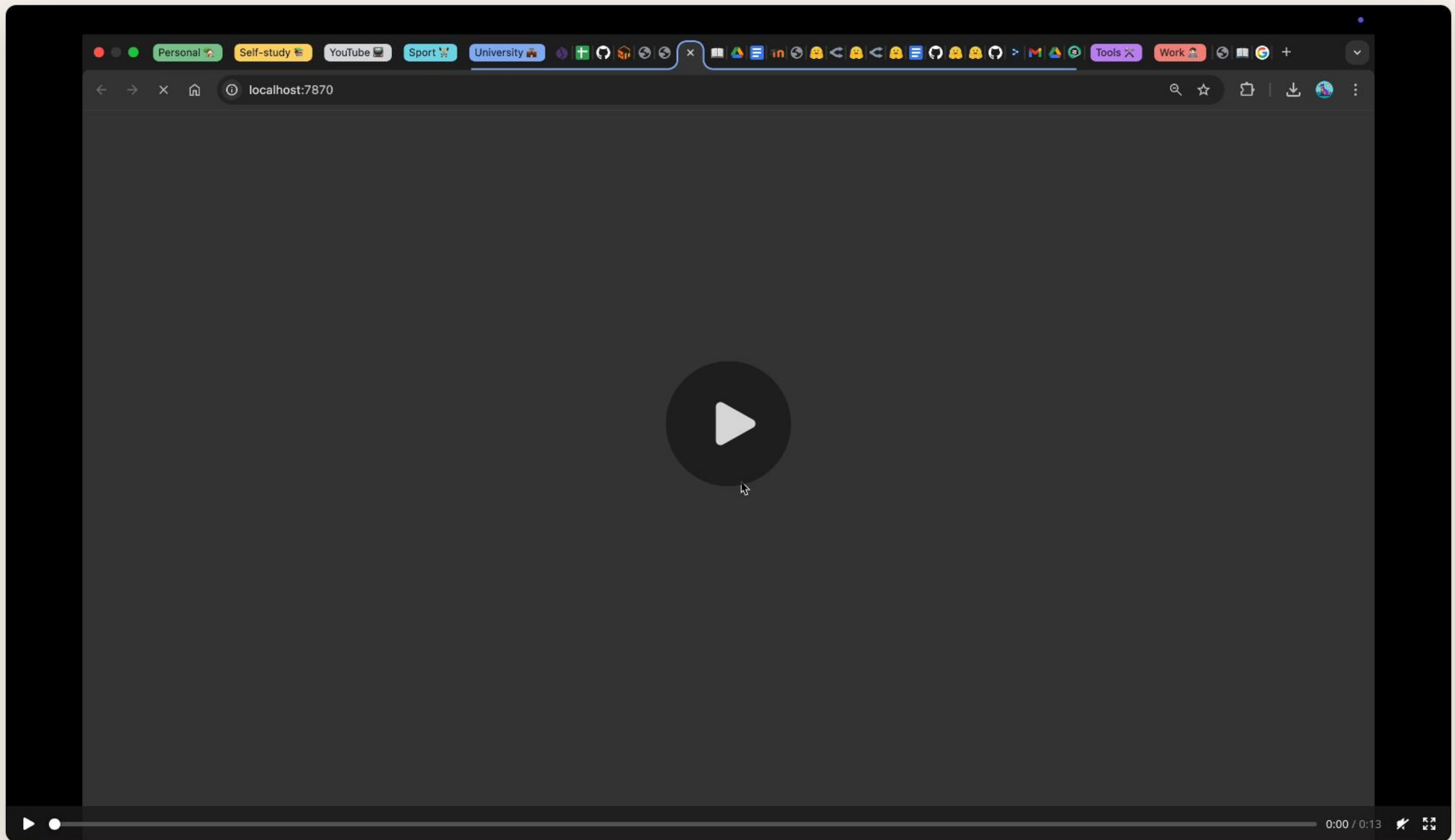
Fine-tuned umt5-base, beam search with 5 beams, float16 on GPU.

Voice input

ASR module speechbrain/m-ctc-t-large; WebM, OGG and WAV are converted to 16 kHz mono.

UX

Four variety presets with descriptions and one example sentence per variety.





— CONCLUSIONS

Conclusions

- 01** Assembled a parallel corpus for surzhyk and three dialects: **125,615 / 13,756 / 200** pairs from two synthesis pipelines (rules with pymorphy2; Mistral-Small-24B). For Boiko and Transcarpathian this appears to be the first public parallel data.
- 02** Fine-tuned two models of different architectures: umt5-base (full fine-tuning) and MamayLM / Gemma-3-4B (QLoRA, 300× fewer trainable weights).
- 03** On surzhyk and Hutsul the fine-tuned models scored above the zero-shot commercial LLMs: **chrF++ 93.58 vs 86.67** and **86.90 vs 65.34**.
- 04** Where training data was synthetic only (Boiko, Transcarpathian), the commercial LLMs stayed ahead; manual annotation quality definitely matters more than data volume.
- 05** Built the «Lexicon» web application: four varieties, normalization with umt5-base and voice input through an ASR module.

93.58

chrF++ on surzhyk (umt5-base)

86.90

chrF++ on Hutsul (umt5-base)

125,615

training pairs in the corpus

4

varieties in one system

Thank you for your attention.